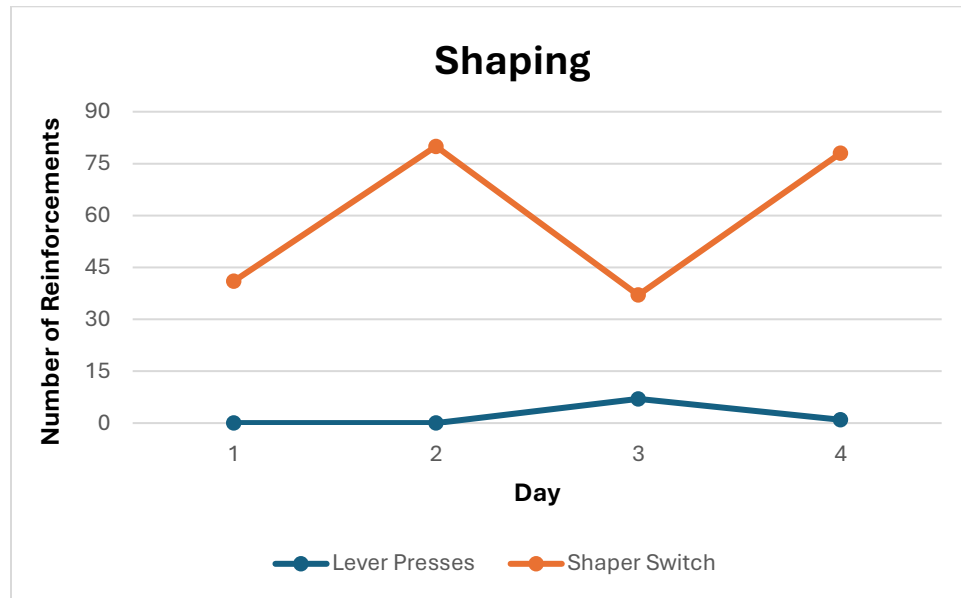


Assignment #1:

(Handling, Magazine Training, and Shaping)

Graph(s):



Short-answer questions:

- In Experiment 1, you were required to handle the rat. What is the goal of handling and how will the results of your experiment be affected if you had not handled the rat prior to the other experiments? Give specific examples.

The goal of handling our rats was to get them acquainted with us, their “handlers,” as rats are highly intelligent and can recognize individuals via one’s unique bodily scent. Performing this first experiment is essential as it allows the rats to, not only get our rats more comfortable with us handling them and exploring the new environment but also serves the purpose of being the first “conditioning” session so that our rat can begin understanding our presence as a “secondary reinforcer (S+)”. Additionally, this session allows us “handlers” to become more comfortable with our rat (the rats can sense when we are relaxed/stressed and is more likely to reflect that same behavior). The third and final goal was to obtain the rat’s initial body weight so that we can detect if the rat’s body weight has dropped below 80% at any point, which would indicate a concern for its health. If we had not done all this, we

would observe an obvious lack in the number of lever presses later on because the rats learn better when they are aware of an outcome and what stimuli/behaviors predict it (after this experiment, the water-deprived rats are now motivated by the presence of us handlers who after this experiment now predict the daily sequence of events which eventually results in the administration of water).

- **In Experiment 2, the goal of this exercise is to classically condition the rat to locate and successfully partake of the reinforcement (water). For magazine training, what is the US, CS, and UR?**

US = water

CS (there are 3)= 1. the sound of the dispenser

2. the sight of the lever (becomes secondary reinforcer)

3. the feel of water licked from dispenser (becomes secondary too)

UR = drinking the water

- **In Experiment 3, you were required to shape your rat's behavior. Explain your methodology for reinforcing successive approximations of lever-pressing behavior, then moving toward non-reinforcement of earlier response forms. What worked and what did not, and why do you think various strategies worked or didn't?**

To shape my rats behavior (eventually creating the specific "CR" of my interest), I would initially administer water (activate the dispenser) when the rat climbed/sniffed the portion of the wall next to the lever (so that it would realize that something with that corner of the box resulted in a reward). Once my rat had learned that this specific corner was associated with a reward, I began moving toward non-reinforcement by reward behaviors which ONLY consisted of touching/sniffing the lever. This worked for a little bit as the number of lever interactions began to increase, but stopped working as the issue arose of the rat not yet being able to distinguish between pressing the lever down with its hands and pushing the lever up with its nose. To resolve this and shape my specific behavior of interest, I now extended non-reinforcement to the behavior of nose-poking the lever. Rather quickly my rat distinguished that lever-pressing results in reward while nose-poking did not. This process worked by the end of the lab session as my rat was only now pressing down on the lever.

Garrett Brownlee
Brooks Brissette
NSC 3120 07
10/03/2025

- **How does magazine training help in shaping?**

Magazine training is critical for effective shaping of the rat's behavior. Performing a magazine training generates within the rat a learning process in which an association is formed between WHERE the administration of water (positive reinforcement) takes place and the sounds occurring with this administration of water. It would be very hard (essentially impossible) to shape the animal's behavior if it does not clearly realize that water has been made available to it (whether this be because it didn't know where water was administered or if it didn't recognize the sound associated with the occurrence).